



Design brief

You are a super talented consultant hired by a real estate developer to help design the neighborhood of the future. The client is very hyped by all things tech and believes that smart buildings and cities are the next big thing in real estate. The project is a neighborhood of 12 single family homes, all equipped with heat pumps, solar photovoltaics, and electric vehicles. Architects have already designed the neighborhood layout, all the homes, and specified the PV systems; the final task is to create the storage and control strategies. Your task is to explore all the ways to apply smart controls, determine the utility of storage, and identify an energy management strategy with the most potential. This strategy should be able to minimize costs in response to power and energy price signals, as well as peak events in the grid and islanding in the event of outages.

The client has no specific performance targets you must hit; in other words, they won't know what they want until they see some results. This is a common situation and requires selecting relevant key performance indicators (KPI) so the client understands the diverse benefits of smart energy management and the costs for each feature. Therefore, the analysis requires tests of each technology and multiple system configurations, so they receive a full picture of their alternatives.

The homes

Being a modern, sustainability-focused development, all the homes are designed with energy efficiency in mind. The envelopes are airtight with lots of insulation, heat recovered ventilation, and in-floor heat distribution. All heating is supplied by air-to-water heat pumps with a storage tank for the domestic hot water. Space heating does not use a tank due to the thermal mass of the concrete floor and the need to save space. The home sizes are all the same, but the architects have cleverly made it easy to rearrange them such that they don't look identical. This means that some homes have their PV systems on south facing roofs, and others have them oriented east/west.

All the homes have already been sold, so the developer knows how many people will be living in each house. It is a very diverse neighborhood, from multi-generational households to young families and single pensioners. Most homes only have one electric vehicle, but some larger families require two EVs. This diversity in ownership will lead to different heating demands and appliance load patterns. At the end of this document is a list describing the common attributes and unique characteristics of each home, which will need to be applied in the simulation models.



Process, requirements, and limitations

The first step in any design is to identify your goals (objectives) and how you are going to measure them. Refer to the lecture materials for inspiration and lean towards simplicity. Use KPIs for informing your conclusions and provide additional context or explanation with secondary indicators in your discussions. Be sure to have indicators that represent the various goals of the neighborhood and its residents, be them technical, economic, and environmental. Since the client does not have any specific goals, the final report needs to communicate the tradeoffs that come with prioritizing one KPI over another, which will inform their own decision-making process. Even though you are not the final decision maker, you should also provide your own recommendations based on the objective results and your own subjective point of view. Just be sure to motivate your opinions with reason and logic.

It is important to keep in mind that there are several stakeholders in the neighborhood that will be impacted by your design; it is not only the developer and residents. Be sure to apply relevant KPIs to each stakeholder you consider in the design and clearly redefine KPIs if and when they are altered by point-of-view or system boundary level. Boundaries of what will be acceptable to a given stakeholder should be taken into account; for example: increasing the cost of energy by 200% for increasing self-sufficiency from 37% to 40% is difficult to see as realistic. There are no hard boundaries given, so when it doubt bring your ideas to the consultations for discussion.

Design of a smart building energy system (E-level analysis)

The first step of the analysis is to identify which technologies and control strategies offer the most interesting results in a smart home energy management strategy. Therefore, the only stakeholder to consider is the one homeowner and the design objectives can be centered around their needs. Controllers for all technologies must be tested to motivate the final recommendations, including specifications for storage capacities. All the homes come with pre-defined equipment, and it is not required to change or add further equipment if it does not contribute to the design objectives.

There are three controller strategies to investigate: **self-consumption** (i.e. maximizing solar energy utilization), **price arbitrage** (i.e. minimizing the cost of grid electricity), and **peak shaving** (both for importing and exporting of electricity). Each approach should be tested **individually** with each technology for nine combinations in total. **Do not stack strategies on top of each other**: this stage is limited to testing each method and technology independently. The following list describes the analysis for each technology.

Heat pumps

Smart heat pump controls are based on utilizing thermal storage to modify how electricity is consumed. Therefore, you should test controls that utilize the domestic hot water tank and the building mass to achieve your design goals. The given model's tank is the minimum necessary and larger sizes should be tested. Maximum size is 700 liters and can be tested in 100-liter increments. As the tank size increases, think about the sensor location and how you might control the volume.

Battery storage

Stationary batteries are the chemical storage alternative (in contrast to thermal) for electricity and are connected to the system via AC. There is a 16 kWh maximum capacity which can be tested in 4 kWh increments, and it is possible to recommend no battery. Specific battery models for the building and community level are recommended below.



Electric vehicles

EV controls are about deciding when to charge and with how much power. You should not consider bi-directional charger strategies where the EV is used for routine discharging into the home or grid.

The smart neighborhood as a collection of smart homes (C-level analysis)

With an understanding of your smart home alternatives, the next step is to create a single controller you will deploy across the neighborhood. You are not required to use all three strategies in the controls of each device, but each device should have at least one smart strategy applied within its controller.

With a single control strategy prepared, you must then apply it to each home in the neighborhood and assess their performance. You can adjust settings within the controllers if needed to meet energy demands, but the general algorithmic strategies must remain constant. The aim is to measure how your controller performs when deployed at scale. Use the same KPIs throughout the analysis to ensure you are measuring performance consistently throughout the neighborhood and study.

The final step is to sum the net loads of all buildings and compare this to the non-smart neighborhood. This is a collection of smart homes, but not necessarily a smart city. The analysis must include a discussion of how well the final load curve meets the goals of the community, which are directly linked to the interests of the local grid company and other nearby neighborhoods looking to electrify. The neighborhood will be connected to the network at a single point and the size of this cable is based on the expected peak loads. Any heterogeneity within the community regarding the relative benefits received from individual homes should be discussed as perceived differences can affect homeowner satisfaction with the system.

Community energy systems (A-level analysis)

With distributed controls understood, centralized community energy system strategies should be tested; where the entire community is one prosumer, meaning they are all behind the meter within a smart microgrid. According to current laws, the microgrid must be privately owned, therefore an additional cost must be included to unlock the greater flexibility potential of the community.

There are two alternatives to test at this scale: 1) a centralized heat pump and micro district heating network means that a more efficient ground source heat pump is used with greater storage capacity, and 2) all distributed batteries are removed and a single, centralized battery is applied within a microgrid. The same general control strategies developed for single homes can be applied to the community level; it is not required to test all alternatives a second time, but storage capacities (tanks and batteries) must be evaluated.

The ability for the community to meet any technical or economic goals relative to the distributed approach should be discussed. Since the households are not modeled individually at this scale, it is necessary to divide the economic costs and benefits for each household. There are several methods that can be used (e.g. equal per household, equal per person, or proportional to energy use) and should be referenced in the discussion. Therefore, a qualitative analysis on how the changes may or may not be attractive for individuals and their willingness to participate is needed.

Finally, the entire analysis should be used to discuss policies and business models which can encourage smarter use of energy in communities. Are the economic incentives aligned with technical and environmental goals? What pricing levels would you recommend to the electric companies? Are there any barriers to adoption and how could they be removed?

Boundary conditions

To reduce inconsistencies between proposals (and to make the project more manageable), all projects must use the supplied boundary conditions defined in this section. Many parameters are set within the Polysun files provided, for example; the location, building construction, load profiles, occupancy schedules, PV systems and grid conditions are all preset. Edits to the file should therefore be focused on adding equipment and controllers.

The pricing model for purchased electricity includes three components: a fixed annual fee, energy price (kr/kWh), and a peak power price (kr/kW). There is also a separate price for excess energy sold back to the grid (kr/kWh). Energy prices are fixed hourly and a single year of buy/sell prices are provided. Peak power price is applied only to the maximum value each year. Other economic boundary conditions include investment lifetime and discounting, which are also included in the table below.

Item	Unit Price
Fixed annual fee	2000 kr/yr
Peak power price	1000 kr/kW/yr
Investment lifetime	20 years
Real discount rate	3%
Nominal discount rate	5%

All the energy system equipment prices, lifetimes, and annual operations and maintenance prices are given in the table below. Most of the equipment is part of the baseline system, meaning that their installation has already been decided. These parameters are provided in the case that relative total life cycle costs for the system will be used and therefore applied to the entire system's price. Marginal changes in TLCC due to storage capacities will rely on the prices for additional tank volume and batteries. Note that storage prices include both a fixed component and a capacity component, for example a battery costs 25 kkr plus 6000 kr for each kWh.

Item	Price	Lifetime	O&M
Heat Pump System (baseline)	200 kkr	20 yrs	2000 kr/yr
Photovoltaics (baseline)	150 kkr	30 yrs	1500 kr/yr
EV charger (baseline)	10 kkr	10 yrs	-
Additional 100l of tank	500 kr/100l	20 yrs	-
Home battery fixed price	25 kkr	Calculated	2000 kr/yr
Home battery marginal price	6000 kr/kWh	Calculated	-
Community heating system	3.2 Mkr	30 yrs	20 kkr/yr
Community power grid	800 kkr	100 yrs	-
Community battery fixed price	60 kkr	Calculated	4000 kr/yr
Community battery marginal price	3000 kr/kWh	Calculated	-

Controllers can be implemented and operated without any cost (aside from any potential efficiency losses due to storage, which is revealed in the simulation results).



Home descriptions

As a master planned community, all homes have the same floor area, low-energy envelopes, and PV system capacities. Differences come in the appliance profiles, hot water demand profiles, internal gains, absences, and use of the EV.

In most cases, the profiles are labeled with the house number, except for the electricity consumption which uses the profile code given in the table below. The baseline model file provided includes a separate system diagram for each house with all profiles pre-loaded, however you may decide to copy your final controller's system diagram and update the boundaries for each house. Be sure to update the profiles in each component; you can also cross reference against the baseline models in case the demands don't match.

The electricity demand is a function of the consumption profile; when selecting a profile, enter "0" as the annual demand and it will be replaced with the actual consumption in the file. On the contrary, the hot water demand is an input and must be adjusted for each house. The final water demand will not match the input exactly due to the absences.

The community was designed to be net-zero, but was planned before the homes were sold/rented. Net-zero was achieved (within 2%), but each home has a notably different solar fraction, ranging from 76% to 145%. The PV systems are 9.0 kW_p with some facing due south and some facing due west. All roofs have a 25° tilt angle.

Due to a great offer provided by the dealer, all homeowners have bought a Tesla Model S 75D as their EV (catalog no. 6). They also have an 11 kW AC Level 2 charger (catalog no. 9). The neighborhood's demographics are quite diverse, so you will find unique use characteristics for each EV that significantly impact their availability to the system.

For batteries it is recommended to select separate models for each capacity, do not use a single battery from the catalogue and add multiples. For individual homes, Sonnen has a good range of capacities available with their Eco models. At the community scale, Tesvolt AG is a good choice. Use the nominal capacity when calculating the battery cost.

Hus No.	Residents	Water (l/day)	Electric (kWh/yr)	PV Gen (kWh/yr)	Self Cons
1001	Young couple, both working	140	9 337	8 203	23%
1002	Mid-age couple, both working	140	6 932	8 699	17%
1003	Mid-age couple, both working, one home	140	9 717	8 699	34%
1004	2 adults, both working, 1 child	210	7 747	8 203	26%
1005	2 adults, both working, 3 children	280	10 637	8 699	30%
1006	2 mid-age adults plus 1 elderly adult	200	7 176	8 699	25%
1007	Couple, aged over 65, retired	80	5 655	8 203	19%
1008	2 adults, both work, one home, 3 children	280	11 515	8 699	38%
1009	Single mother, 1 child, working	130	6 186	8 203	21%
1010	Single man over 65, retired	60	6 186	8 203	19%
1011	Single working father, 2 children	140	7 540	8 699	21%
1012	2 adults, 2 children, one parent works	225	10 858	8 203	38%
Total	Entire Community	-	99 651	101 412	26%



Project and modeling tips

Like most things in energy development and engineering, there are many grey areas in this project that you must navigate, and the solutions won't always be so clear. It is highly encouraged to ask questions about any aspect of the project, but they should be questions about ideas, approaches, or results rather than "can you show me how to do this." The project should be a dialog about how we can find the best possible set of tradeoffs from your perspective.

That does not mean you won't be able to ask questions about the modeling or Polysun, that would be crazy. Every effort has been made to provide the resources necessary for the project, so start there and ask if something is missing. It is entirely possible that something has been overlooked (or likely...there are a lot of parameters) or an error made in the supplied files, so if something just doesn't look right or work right please bring it up.

Polysun as a simulation tool has its limitations and quirks (like any tool) but its greatest strength is how approachable it is. One specific feature that will be useful for you is the library of included templates, which provide working examples of various controllers. Of particular interest will be the 50-series which are all about grid-connected PV with various battery controllers and EVs. Use these templates as inspiration and guides in your own controller designs.